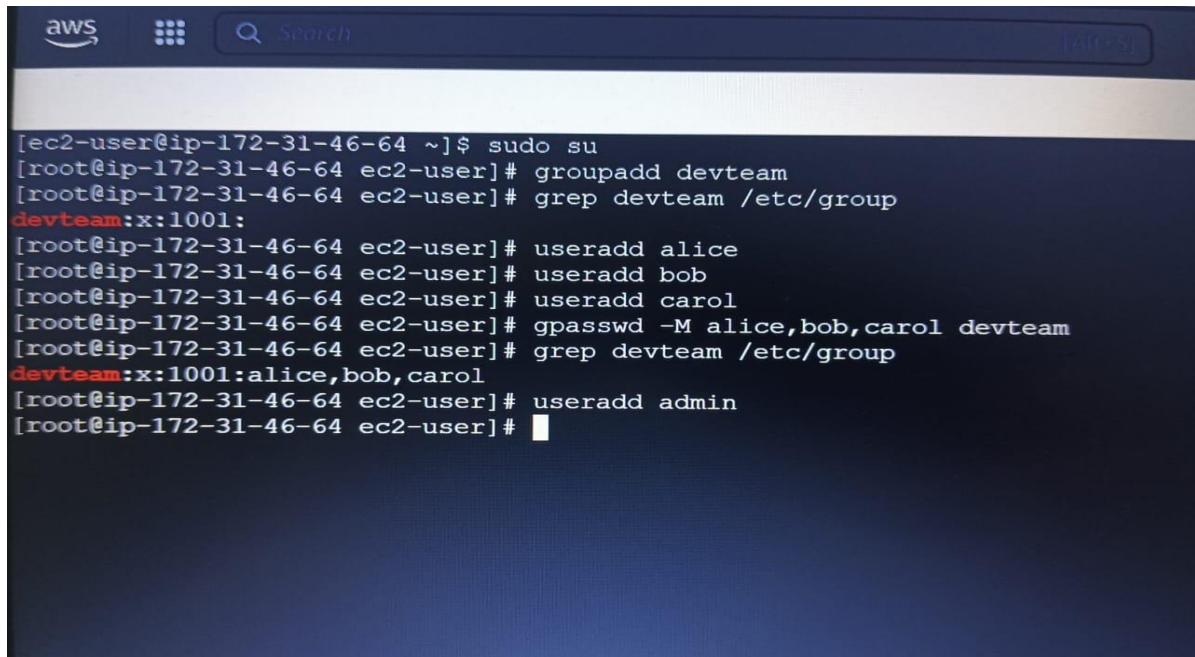


1. User and Group Management

- Create a group named devteam.
- Create three users: alice, bob, and carol.
- Add all three users to the devteam group.
- Create a user named admin who is **not** part of the devteam group.



```
aws [Search] [logout]
[ec2-user@ip-172-31-46-64 ~]$ sudo su
[root@ip-172-31-46-64 ec2-user]# groupadd devteam
[root@ip-172-31-46-64 ec2-user]# grep devteam /etc/group
devteam:x:1001:
[root@ip-172-31-46-64 ec2-user]# useradd alice
[root@ip-172-31-46-64 ec2-user]# useradd bob
[root@ip-172-31-46-64 ec2-user]# useradd carol
[root@ip-172-31-46-64 ec2-user]# gpasswd -M alice,bob,carol devteam
[root@ip-172-31-46-64 ec2-user]# grep devteam /etc/group
devteam:x:1001:alice,bob,carol
[root@ip-172-31-46-64 ec2-user]# useradd admin
[root@ip-172-31-46-64 ec2-user]#
```

2. File Permissions

- Create a directory named /project/team1.
- Set the group ownership of this directory to devteam.
- Set permissions so only members of devteam can read, write, and execute files in the directory.
- Ensure that other users (including admin) do **not** have access.

```

[root@ip-172-31-20-122 ec2-user]# mkdir -p /project/team1
[root@ip-172-31-20-122 ec2-user]# chown :devteam /project/team1
[root@ip-172-31-20-122 ec2-user]# ls -ld /project/team1
drwxr-xr-x. 2 root devteam 6 Nov 22 08:38 /project/team1
[root@ip-172-31-20-122 ec2-user]# chmod 770 /project/team1
[root@ip-172-31-20-122 ec2-user]# ls -ld /project/team1
drwxrwx---. 2 root devteam 6 Nov 22 08:38 /project/team1
[root@ip-172-31-20-122 ec2-user]# su bob
[ bob@ip-172-31-20-122 ec2-user]$ cd /project/team1
[ bob@ip-172-31-20-122 team1]$ exit
exit
[root@ip-172-31-20-122 ec2-user]# su admin
[ admin@ip-172-31-20-122 ec2-user]$ cd /project/team1
bash: cd: /project/team1: Permission denied
[ admin@ip-172-31-20-122 ec2-user]$

```

3. Access Control List (ACL)

- Grant read-only access to the /project/team1 directory for the admin user using ACL.
- Verify and display the ACL settings for this directory.

```

[ admin@ip-172-31-20-122 ec2-user]$ exit
exit
[root@ip-172-31-20-122 ec2-user]# setfacl -m u:admin:r /project/team1
[root@ip-172-31-20-122 ec2-user]# getfacl /project/team1
getfacl: Removing leading '/' from absolute path names
# file: project/team1
# owner: root
# group: devteam
user::rwx
user:admin:r--
group::rwx
mask::rwx
other::---

```

I

```

[root@ip-172-31-20-122 ec2-user]#

```

4. Tar and Gzip

- Create a compressed (tar + gzip) backup of the /project/team1 directory.
- Name the backup file team1_backup.tar.gz and store it in /tmp.

```
[root@ip-172-31-20-122 ec2-user]# tar -cvf /tmp/team1_backup.tar.gz /project/team1
tar: Removing leading `/' from member names
/project/team1/
/project/team1/f1
[root@ip-172-31-20-122 ec2-user]# /tmp
bash: /tmp: Is a directory
[root@ip-172-31-20-122 ec2-user]# cd /tmp
[root@ip-172-31-20-122 tmp]# ls
systemd-private-9df84d89c8614739a8455a61fe9aa9f2-chrond.service-R18fQ7
systemd-private-9df84d89c8614739a8455a61fe9aa9f2-dbus-broker.service-jbQC0S
systemd-private-9df84d89c8614739a8455a61fe9aa9f2-policy-rottes@ens5.service-QSmsq3
systemd-private-9df84d89c8614739a8455a61fe9aa9f2-systemd-logind.service-V3NXk5
systemd-private-9df84d89c8614739a8455a61fe9aa9f2-systemd-resolved.service-1Vr6b0
team1_backup.tar.gz
[root@ip-172-31-20-122 tmp]# du -h team1_backup.tar.gz
12K    team1_backup.tar.gz
[root@ip-172-31-20-122 tmp]#
```